

PHYSICAL AND MAC LAYERS

MOST IMPORTANT UNIVERSITY QUESTIONS (SPPU)

2 MARKS QUESTIONS

1. Define Physical layer.
2. List any four functions of Physical layer.
3. Define MAC layer.
4. List any four functions of MAC layer.
5. What is the role of PHY in data transmission?
6. What is data rate?
7. What is modulation?
8. Expand CSMA/CA.

4 MARKS QUESTIONS

1. Explain functions of Physical layer.
2. Explain the services provided by MAC layer.
3. Explain CSMA/CA mechanism.
4. Explain beacon-enabled mode.
5. Explain non-beacon mode.
6. Differentiate between Physical and MAC layers.

6 MARKS QUESTIONS

1. Explain Physical layer in detail.
2. Explain MAC layer in detail.
3. Explain CSMA/CA with neat diagram.
4. Explain beacon-enabled and non-beacon modes.
5. Compare Physical and MAC layers in detail.

8 MARKS QUESTIONS

1. Explain Physical layer with neat diagram.
2. Explain MAC layer architecture and services in detail.
3. Describe CSMA/CA mechanism with examples.
4. Explain the operation of MAC layer in IEEE 802.15.4.
5. Discuss the interaction between Physical and MAC layers.

SHORT NOTES (4 MARKS)

1. Physical Layer
2. Data Rate
3. Modulation
4. MAC Layer
5. CSMA/CA
6. Beacon Enabled Mode
7. Non-Beacon Mode
8. GTS (Guaranteed Time Slot)
9. Superframe Structure
10. Frame Types

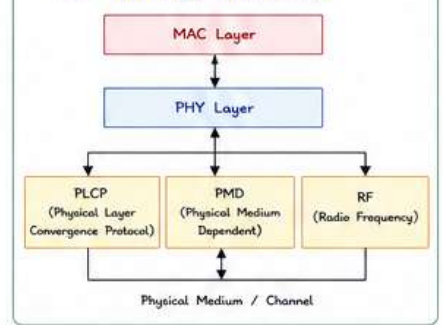
1. PHYSICAL (PHY) LAYER

Physical layer is the lowest layer in the protocol stack. It is responsible for transmitting raw bits over the physical medium.

1.1 Functions of Physical Layer

- Activation and deactivation of radio transceiver.
- Energy detection (ED).
- Clear Channel Assessment (CCA).
- Link quality indication.
- Channel selection.
- Transmission and reception of bits.
- Modulation / Demodulation.
- Data rate selection.

1.2 PHY Layer Architecture



1.3 PHY Layer Parameters (IEEE 802.15.4)

Parameter	Value / Description
Frequency Bands	868-868.6 MHz (Europe) 902-928 MHz (USA) 2400-2483.5 MHz (Worldwide)
Data Rates	20 kbps (868 MHz) 40 kbps (902-928 MHz) 250 kbps (2.4 GHz)
Modulation	BPSK (868/915 MHz) O-QPSK (2.4 GHz)
Channels	1 channel (868 MHz) 10 channels (915 MHz) 16 channels (2.4 GHz)
Receiver Sensitivity	-85 dBm (typical)
Output Power	Up to +20 dBm (typical)

1.4 Modulation Techniques

Band	Modulation
868 MHz	BPSK
915 MHz	BPSK
2.4 GHz	O-QPSK

Role of PHY Layer

It converts digital bits from MAC layer into signals suitable for transmission over the wireless medium and vice versa.

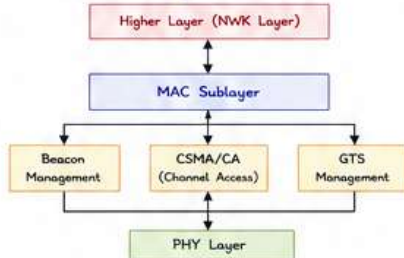
2. MAC (MEDIUM ACCESS CONTROL) LAYER

MAC layer provides reliable data transfer between devices sharing the same medium.

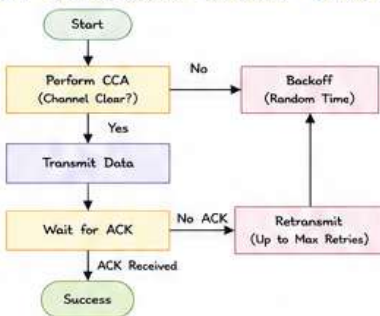
2.1 Functions of MAC Layer

- Beacon management.
- Superframe structure management.
- Channel access mechanism (CSMA/CA).
- GTS management.
- Frame validation.
- Acknowledgement handling.
- Data protection and security.
- Association and disassociation.

2.2 MAC Layer Architecture



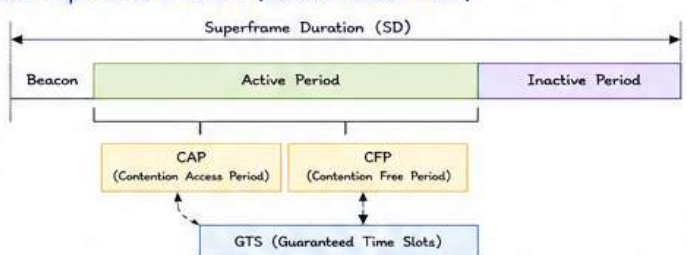
2.4 Channel Access Mechanism – CSMA/CA



Steps in CSMA/CA

1. Sense the channel (CCA).
2. If channel is idle $\rightarrow$ transmit.
3. If channel is busy $\rightarrow$ wait for random backoff time.
4. Transmit data.
5. Wait for ACK.
6. If ACK not received $\rightarrow$ retry (up to maximum retries).
7. If ACK received $\rightarrow$ success.

2.5 Superframe Structure (Beacon Enabled Mode)

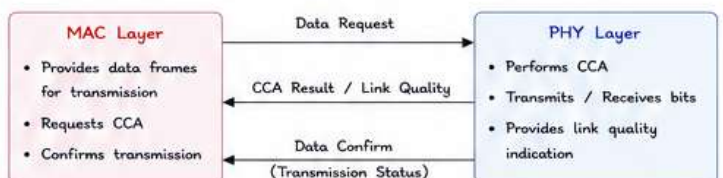


In non-beacon mode, there is no superframe. Devices use CSMA/CA for all transmissions.

3. PHYSICAL LAYER vs MAC LAYER

Feature	Physical Layer	MAC Layer
Position in Stack	Below MAC layer	Above Physical layer
Main Responsibility	Bit transmission over medium	Reliable data transfer between devices
Data Unit	Bits	Frames (MPDU)
Functions	Modulation, Demodulation, CCA, Link Quality	CSMA/CA, Beacon, GTS, Acknowledgement, Security
Addressing	Not required	Uses 16-bit / 64-bit addresses
Connection	Not connection oriented	Supports association
Example	Radio, Transceiver	802.15.4 MAC

4. INTERACTION BETWEEN PHY AND MAC LAYERS



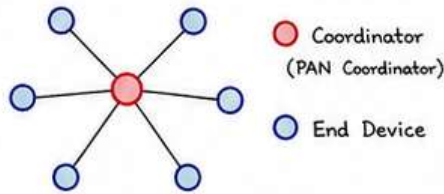
CONCLUSION : Physical layer deals with transmission of raw bits over the wireless medium, while MAC layer ensures reliable and efficient communication through channel access, framing, acknowledgements and security. Together, they provide a robust foundation for IEEE 802.15.4 based IoT networks.

Q. Classify the different Topology of IEEE 802.15.4 and explain with suitable diagram. Also give applications. [6 Marks]

Introduction : IEEE 802.15.4 defines physical layer and MAC layer for Low Rate Wireless Personal Area Networks (LR-WPANs). It supports five types of network topologies to meet different application needs.

Topologies of IEEE 802.15.4

1) Star Topology

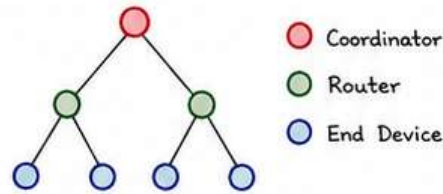


- All devices communicate with central coordinator.
- No direct communication between end devices.
- Simple and low power.

Applications :

- Home automation
- Smart lighting
- Industrial monitoring
- Personal Area Networks

2) Tree Topology

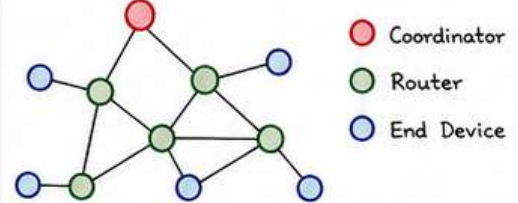


- Hierarchical structure.
- Data flows from end device to coordinator through routers.
- Supports more devices than star.

Applications :

- Smart buildings
- Meter reading systems
- Environmental monitoring
- Agricultural monitoring

3) Mesh Topology

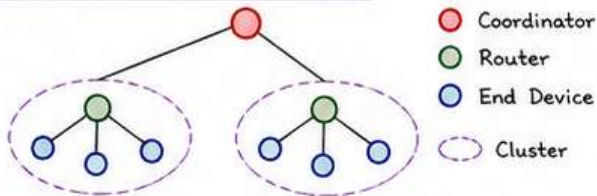


- Devices (routers) can communicate with each other.
- Multiple paths available for data.
- Highly reliable and self-healing.

Applications :

- Smart city networks
- Industrial automation
- Healthcare monitoring
- Large scale sensor networks

4) Cluster Tree Topology

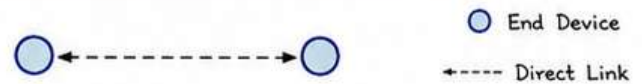


- Combination of cluster and tree.
- Each cluster has a router and end devices.
- Better scalability and management.

Applications :

- Large scale industrial networks
- Smart energy grid
- Building automation systems

5) Peer-to-Peer (Link) Topology



- Two devices communicate directly without any coordinator or router.
- Very simple and low overhead.
- Suitable for point-to-point communication.

Applications :

- Device pairing (e.g., remote control)
- Wearable devices
- Simple sensor to actuator communication

Legend : ● Coordinator (PAN Coordinator) ● Router ● End Device ----- Communication Link

Comparison	Star	Tree	Mesh	Cluster Tree	Peer-to-Peer
Structure	Centralized	Hierarchical	Distributed	Cluster + Hierarchical	Point-to-Point
Complexity	Low	Medium	High	High	Very Low
Reliability	Low	Medium	High	High	Low
Power Consumption	Low	Medium	Medium	Medium	Very Low
Scalability	Low	Medium	High	Very High	Low
Use Case	Small Networks	Medium Networks	Large Networks	Very Large Networks	Simple Link

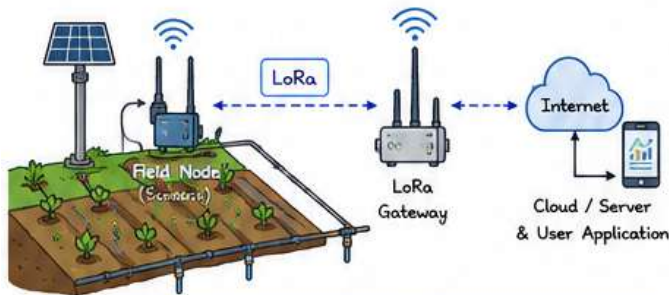
Conclusion : IEEE 802.15.4 supports Star, Tree, Mesh, Cluster Tree and Peer-to-Peer topologies. Selection of topology depends on application requirements like range, reliability, scalability and power.

Q. Show the use of LoRa protocol in the smart irrigation system development. [5 Marks]

Introduction : LoRa (Long Range) is a LPWAN communication protocol that enables low power, long range and reliable data transmission. It is ideal for smart irrigation where sensors send data from farms to a centralized system for monitoring and control.

1) Smart Irrigation System Overview

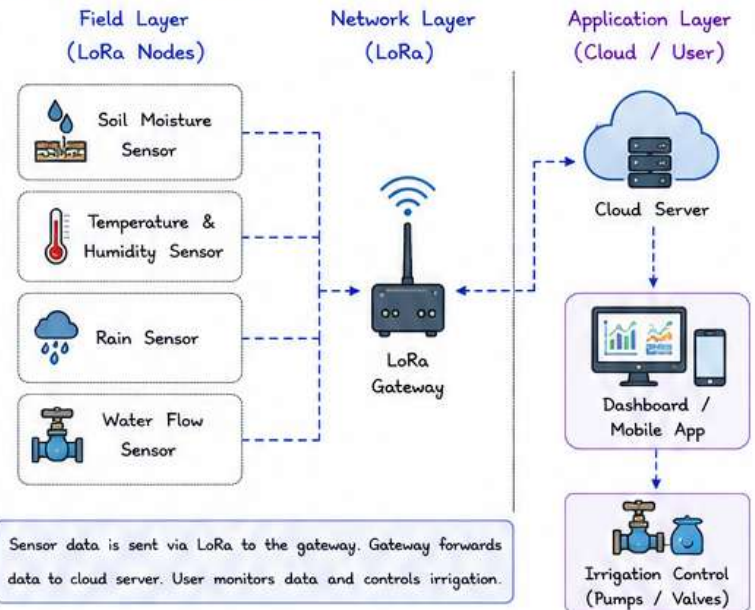
Smart irrigation uses sensors to monitor field conditions and LoRa to send data to the gateway. Based on the data, water pumps/valves are controlled automatically or manually for efficient water usage.



Benefits in Irrigation :

- Saves water
- Improves crop yield
- Saves energy
- Enables remote monitoring and control

2) System Architecture



3) How LoRa is Used in Smart Irrigation

- Sensors in the field (soil moisture, temperature, flow, rain) collect real-time data.
- LoRa enables long range communication (2-15 km in rural areas) with very low power.
- Data is sent from LoRa nodes to LoRa gateway.
- Gateway forwards data to cloud/server using Wi-Fi/4G/Ethernet.
- User dashboard shows field data and irrigation status.
- Based on data, system automatically turns ON/OFF the pump or valve, or user can control manually.

Why LoRa?

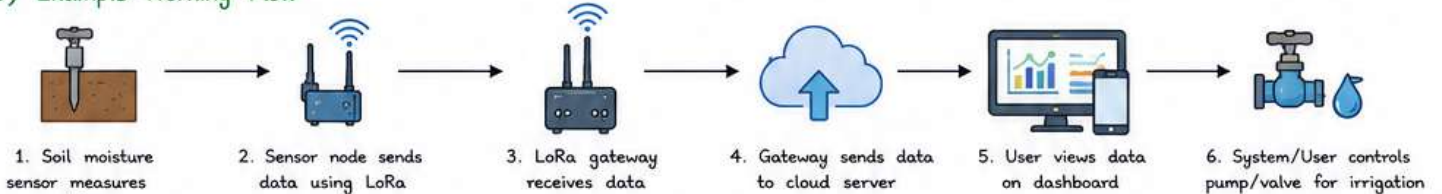
- ✓ Long range - Covers large agricultural fields
- ✓ Low power - Battery operated nodes last for years
- ✓ Low cost - Suitable for wide area deployment
- ✓ High penetration - Works well in rural areas
- ✓ Supports many nodes - One gateway handles many sensors

4) LoRa Protocol Features Used

Feature	Description
Modulation	Chirp Spread Spectrum (CSS) - robust to interference
Frequency Bands	Sub-GHz ISM bands (e.g., 868 MHz / 915 MHz)
Range	2 to 15 km (rural), up to 5 km (urban)
Data Rate	0.3 kbps to 50 kbps (adaptive)
Payload Size	Up to 222 bytes per packet
Power Consumption	Very low - suitable for battery devices
Network Topology	Star-of-stars (Nodes → Gateway → Server)
Security	AES-128 encryption for secure data

These features make LoRa ideal for smart irrigation where data rate requirement is low but range and battery life are critical.

5) Example Working Flow



Example : If soil moisture is low → System opens the valve / turns ON the pump → Water is supplied → When moisture is sufficient → Valve closes / pump OFF automatically.

6) Advantages in Smart Irrigation

- Real-time monitoring of field conditions
- Optimized water usage based on actual need
- Reduces manpower and operational cost
- Early detection of issues (dry soil, leak, pump failure)
- Scalable system - can add many fields/nodes easily
- Works in remote areas without internet at field level

Applications

- Agriculture farms
- Greenhouses
- Orchards
- Golf courses
- Landscaping
- Community irrigation projects

Conclusion

LoRa protocol enables long range, low power and reliable communication between field sensors and central system. It is well suited for smart irrigation systems to improve water management, reduce cost and increase crop productivity.

In short : LoRa helps in connecting field sensors over long distance with very low power, making smart irrigation systems efficient, reliable and easy to deploy.

NETWORK LAYER – IP VERSIONS 4 & 6

MOST IMPORTANT UNIVERSITY QUESTIONS (SPPU)

2 MARKS QUESTIONS

1. Define IP (Internet Protocol).
2. List any two differences between IPv4 and IPv6.
3. What is the size of an IPv4 address?
4. What is the size of an IPv6 address?
5. Expand IPv4 and IPv6.
6. What is NAT?
7. What is the purpose of Flow Label in IPv6?
8. What is SLAAC?

4 MARKS QUESTIONS

1. Explain IPv4 address format with example.
2. Explain IPv6 address format with example.
3. Explain the header format of IPv4.
4. Explain the header format of IPv6.
5. Write short notes on NAT.
6. Explain any four differences between IPv4 and IPv6.

6 MARKS QUESTIONS

1. Explain IPv4 and its header format in detail.
2. Explain IPv6 and its header format in detail.
3. Compare IPv4 and IPv6 in detail.
4. Explain the transition mechanisms from IPv4 to IPv6.
5. Discuss the advantages of IPv6 over IPv4.

8 MARKS QUESTIONS

1. Compare IPv4 and IPv6 with suitable diagram.
2. Explain IPv6 addressing, header format and transition mechanisms.
3. Discuss in detail the need for IPv6.
4. Explain various address types in IPv6 with examples.
5. Write a detailed note on NAT and its limitations.

SHORT NOTES (4 MARKS)

1. IPv4 Addressing
2. IPv4 Header
3. IPv6 Addressing
4. IPv6 Header
5. NAT
6. CIDR
7. SLAAC
8. Flow Label
9. Transition Mechanisms
10. Anycast Address

4. IPV4 VS IPV6 COMPARISON

Feature	IPv4	IPv6
Address Size	32 bits	128 bits
Address Notation	Dotted decimal	Hexadecimal (colon separated)
Address Space	4.3×10^9	3.4×10^{38}
Header Size	20–60 bytes	40 bytes (fixed)
Checksum	Present	Not present
Fragmentation	By sender & router	By sender only
Broadcast	Supported	Not supported (uses multicast)
NAT	Required	Not required
Auto Configuration	Manual / DHCP	SLAAC / DHCPv6
Security	Optional (IPsec)	Built-in (IPsec mandatory)
QoS Support	ToS field	Traffic Class + Flow Label

1. INTRODUCTION

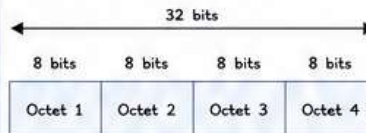
The Internet Protocol (IP) is a connectionless, best-effort protocol that provides logical addressing and routing of packets across interconnected networks.

Two versions of IP are in use:

- IPv4 (Internet Protocol version 4)
- IPv6 (Internet Protocol version 6)

2. IPV4

2.1 IPv4 Address Format



Example: 192.168.1.10

2.3 Key Features of IPv4

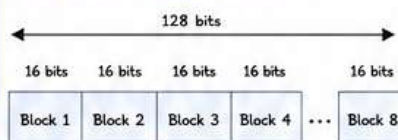
- 32-bit addressing (about 4.3 billion addresses)
- Dotted decimal notation
- Supports unicast, broadcast and multicast
- Uses NAT to overcome address shortage
- Header size: 20 to 60 bytes

2.2 IPv4 Header Format

Version (4)	IHL (4)	Type of Service (8)	Total Length (16)
Identification (16)		Flags (3)	Fragment Offset (13)
Time to Live (8)	Protocol (8)		Header Checksum (16)
Source IP Address (32)			
Destination IP Address (32)			
Options (0 – 40 bytes)		Padding	
Data			

3. IPV6

3.1 IPv6 Address Format



Example:

2001:0db8:85a3:0000:0000:8a2e:0370:7334
(8 blocks of 16 bits, written in hexadecimal)

3.2 IPv6 Header Format

Version (4)	Traffic Class (8)	Flow Label (20)	
Payload Length (16)		Next Header (8)	Hop Limit (8)
Source Address (128)			
Destination Address (128)			
Extension Headers (if any)		Data	

3.3 Key Features of IPv6

- 128-bit addressing (about 3.4×10^{38} addresses)
- Hexadecimal colon-separated notation
- Supports unicast, multicast and anycast
- No NAT required
- Header size: 40 bytes (fixed)
- Built-in security (IPsec support)
- Autoconfiguration (SLAAC)
- Better support for QoS (Flow Label)

5. TRANSITION MECHANISMS FROM IPV4 TO IPV6

1. Dual Stack : Device runs both IPv4 and IPv6.
2. Tunneling : IPv6 packets are carried over IPv4 network.
3. Translation : Protocol translation between IPv4 and IPv6.
4. Header Translation : Converts headers from one version to another.

6. ADDRESS TYPES IN IPV6

1. Unicast : One-to-one communication.
2. Multicast : One-to-many communication.
3. Anycast : One-to-nearest communication.
4. Link-Local : Used within the local link (FE80::/10).
5. Global Unicast : Routable on the internet (2000::/3).

7. NEED FOR IPV6

- Address exhaustion in IPv4.
- Growth of Internet and IoT devices.
- End-to-end connectivity without NAT.
- Better security and QoS support.
- Auto configuration and simplified management.

8. LIMITATIONS OF IPV4 & SOLUTION IN IPV6

Limitation in IPv4	Solution in IPv6
Limited address space	128-bit address space
Requires NAT	End-to-end connectivity (No NAT)
Header checksum processing	Checksum removed from header
Broadcast overhead	No broadcast, uses multicast/anycast
Complex configuration	SLAAC and DHCPv6 for easy configuration
Weak security	IPsec integration in IPv6

CONCLUSION : IPv6 is the future of internet as it provides vast address space, better security, simplified header and improved support for modern applications like IoT, cloud computing and real-time services.

Q. What is 6LoWPAN? 6LoWPAN and EPC standardization.

[6 Marks]

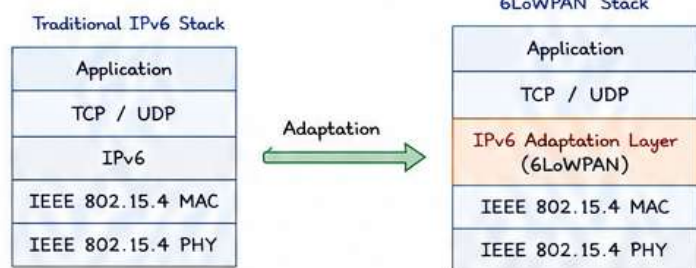
1) What is 6LoWPAN?

6LoWPAN (IPv6 over Low Power Wireless Personal Area Networks) is an IETF standard (RFC 4944) that enables IPv6 communication over IEEE 802.15.4 based low power wireless networks.

Need of 6LoWPAN

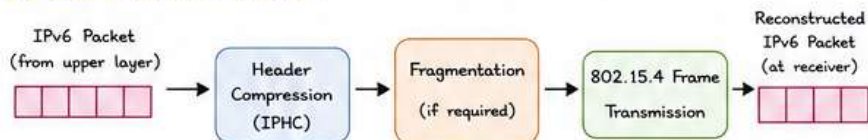
- IEEE 802.15.4 has small frame size (127 bytes).
- IPv6 packet is large (40 bytes header minimum).
- Direct transmission is inefficient.
- 6LoWPAN adapts IPv6 to run efficiently over low power networks.

2) 6LoWPAN Protocol Stack



6LoWPAN adaptation layer compresses and fragments IPv6 packets to fit in small 802.15.4 frames.

3) How 6LoWPAN Works?

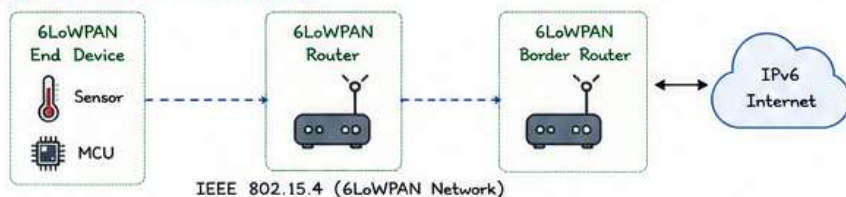


- Header Compression (IPHC) reduces the IPv6 header from 40 bytes to as low as 2-7 bytes.
- Fragmentation splits large packets into smaller pieces.
- Reassembly at the receiver reconstructs the original IPv6 packet.
- Result: Efficient use of bandwidth, low power and small memory.

4) Features of 6LoWPAN

- Enables IPv6 over IEEE 802.15.4 networks.
- Header compression (IPHC) to reduce overhead.
- Fragmentation and reassembly of packets.
- Supports mesh topology.
- Plug and play - uses standard IPv6.
- Supports neighbor discovery and auto-configuration.
- Designed for low power, lossy and constrained environments.

5) 6LoWPAN Architecture



- **End Devices** : Sensors/actuators with limited resources.
- **6LoWPAN Router** : Forwards packets and forms mesh network.
- **Border Router** : Connects 6LoWPAN network to IPv6 Internet.

6) Advantages

- Provides end-to-end IPv6 connectivity.
- Low power consumption.
- Works on low data rate and lossy networks.
- Interoperable with existing Internet applications.
- Scalable for large IoT deployments.
- Cost effective.

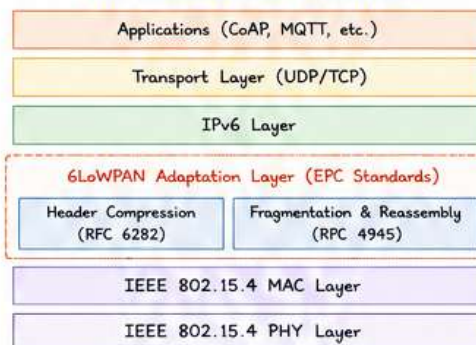
7) EPC Standardization

EPC (Efficient Performance with IPv6 over Low power Wireless Personal Area Networks) is a working group in IETF which defines standards for 6LoWPAN.

EPC Working Group - Key Standards

Standard (RFC)	Description
RFC 4944	Overview and architecture of 6LoWPAN.
RFC 6282	IPv6 Header Compression (IPHC).
RFC 4945	Fragmentation and Reassembly.
RFC 6775	Neighbor Discovery Optimization.
RFC 6775	6LoWPAN Mesh Addressing.
RFC 8505	6LoWPAN over IEEE 802.15.4 - Best Current Practices.
RFC 8180	6LoWPAN over Other IEEE 802.15.4 PHYs.
RFC 8166	RPL: IPv6 Routing Protocol for Low-Power and Lossy Networks (used with 6LoWPAN).

EPC Standardization Architecture



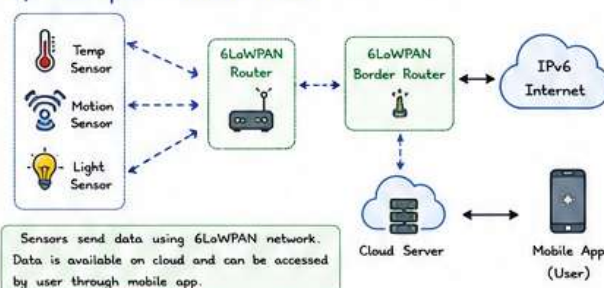
EPC Focus Areas

- ✓ Lightweight IPv6
- ✓ Header Compression
- ✓ Efficient Fragmentation
- ✓ Mesh Addressing
- ✓ Routing (RPL)
- ✓ Interoperability
- ✓ Best Practices

8) Applications of 6LoWPAN

- Smart Home Automation
- Industrial Monitoring and Control
- Smart Energy / Smart Metering
- Environmental Monitoring
- Healthcare and Wearables
- Smart Agriculture
- Building Automation

9) Example - Smart Home



10) Summary

6LoWPAN enables IPv6 communication in low power and constrained networks by compressing headers and fragmenting packets. EPC working group of IETF defines a set of standards that make 6LoWPAN efficient, interoperable and suitable for large scale IoT applications.

Conclusion : 6LoWPAN is the key enabler for bringing IPv6 to resource constrained IoT devices. EPC standardization ensures efficiency, interoperability and scalability for future IoT networks.

Q. What is CoAP? Explain Constrained Application Protocol in detail. [6 Marks]

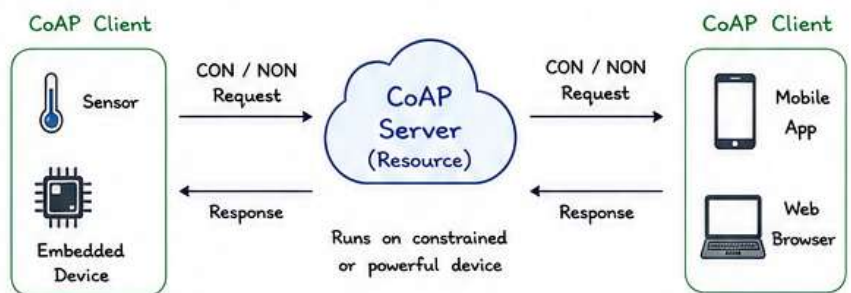
1) What is CoAP ?

CoAP (Constrained Application Protocol) is a lightweight application layer protocol designed for constrained devices and low power, lossy networks such as IoT.

Key Points :

- Developed by IETF (RFC 7252).
- Works over UDP.
- Uses RESTful architecture similar to HTTP but much lighter.
- Ideal for Machine to Machine (M2M) and IoT communication.

2) CoAP Architecture

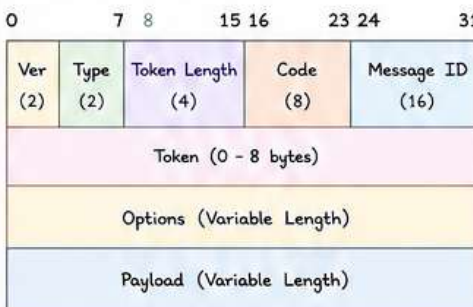


CoAP is a client-server protocol where clients request or modify resources identified by a URI, and server responds.

3) Features of CoAP

- Lightweight: very small header (4 bytes).
- Built for constrained devices.
- Works over UDP (no connection overhead).
- Supports confirmable (CON) and non-confirmable (NON) messages.
- Reliability through message ID, Acknowledgement (ACK) and retransmission.
- Supports Observe option (server push).
- RESTful: Use of methods like GET, POST, PUT, DELETE.
- Easy mapping to HTTP for Web integration.

4) CoAP Message Format



- Ver : Version (always 1)
- Type : CON(0), NON(1), ACK(2), RST(3)
- Token Length : 0 - 8 bytes
- Code : Request/Response code
- Message ID : For matching messages
- Options : URI, Observe, Content-Format etc.
- Payload : Actual data

5) CoAP Methods (Similar to HTTP)

Code (Dec)	Method	Description
1	GET	Retrieve a resource
2	POST	Create a new resource
3	PUT	Update/Replace a resource
4	DELETE	Delete a resource

CoAP Response Codes (like HTTP):

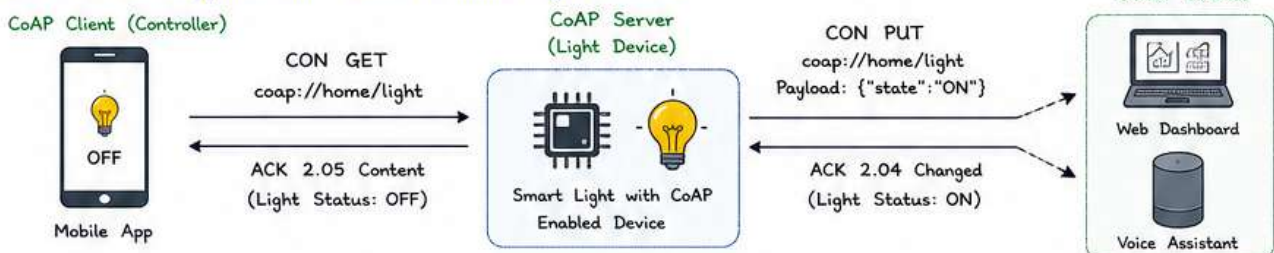
2.01 Created, 2.02 Deleted, 2.05 Content, 4.00 Bad Request, 4.04 Not Found, 5.00 Internal Server Error etc.

6) Message Types in CoAP

Type	Value	Description
Confirmable (CON)	0	Requires Acknowledgement (ACK).
Non-Confirmable (NON)	1	No Acknowledgement. Best-effort delivery.
Acknowledgement (ACK)	2	Sent in response to CON message.
Reset (RST)	3	Indicates problem in receiving the message.

- CON - used when reliability is required.
- NON - used when occasional loss is acceptable.
- ACK - acknowledges CON message.
- RST - sent when message cannot be processed.

7) CoAP in IoT Application - Smart Home Light Control



The mobile app (CoAP client) sends request to control the light. CoAP server (smart light) processes the request and responds. Other clients can also access the same resource.

8) Advantages of CoAP

- Very low overhead (saves bandwidth).
- Suitable for low power devices.
- Supports unreliable networks.
- Simple and easy to implement.
- Interoperable with HTTP and Web.
- Ideal for large scale IoT deployments.

9) Applications of CoAP

- Smart Home Automation
- Building Automation
- Industrial IoT (IIoT)
- Environmental Monitoring
- Smart Agriculture
- Smart Cities
- Wearables and Health Care

10) CoAP vs HTTP

Feature	CoAP	HTTP
Protocol	Application Layer	Application Layer
Transport	UDP	TCP
Overhead	Very Low (4 bytes hdr)	Higher (20+ bytes hdr)
Devices	Constrained Devices	General Purpose Devices
Power Usage	Very Low	Higher
Reliability	Built-in (CON/ACK)	TCP handles
Best For	IoT, M2M	Web Browsing

Conclusion : CoAP is a lightweight, efficient and reliable protocol designed for IoT and constrained environments. It enables simple RESTful communication with low overhead over UDP.

Q. Illustrate the use of MQTT protocol with suitable IoT Application. [6 Marks]

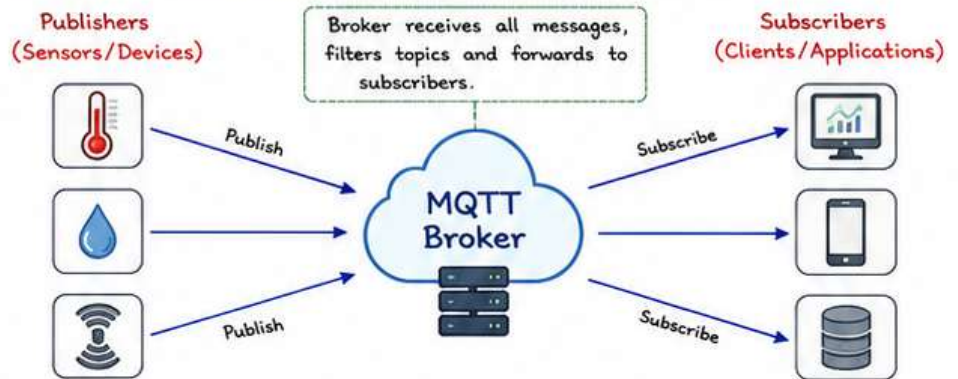
1) What is MQTT ?

MQTT (Message Queuing Telemetry Transport) is a lightweight **publish/subscribe** messaging protocol. It is designed for **low bandwidth**, **unreliable networks** and **resource constrained devices**. It works on top of **TCP/IP**.

2) Features of MQTT

- Lightweight and simple
- Publish / Subscribe model
- Low bandwidth and low power consumption
- Works on unreliable networks
- Quality of Service (QoS) support
- Retained messages
- Last Will and Testament (LWT)
- Secure (supports TLS/SSL)
- Runs on TCP/IP (default port 1883, secure port 8883)

3) MQTT Architecture



4) How MQTT Works ?

- Publishers publish messages to a topic.
- MQTT Broker receives the messages.
- Subscribers subscribe to topics.
- Broker forwards messages of the subscribed topics to the subscribers.
- This is a **publish/subscribe** communication.

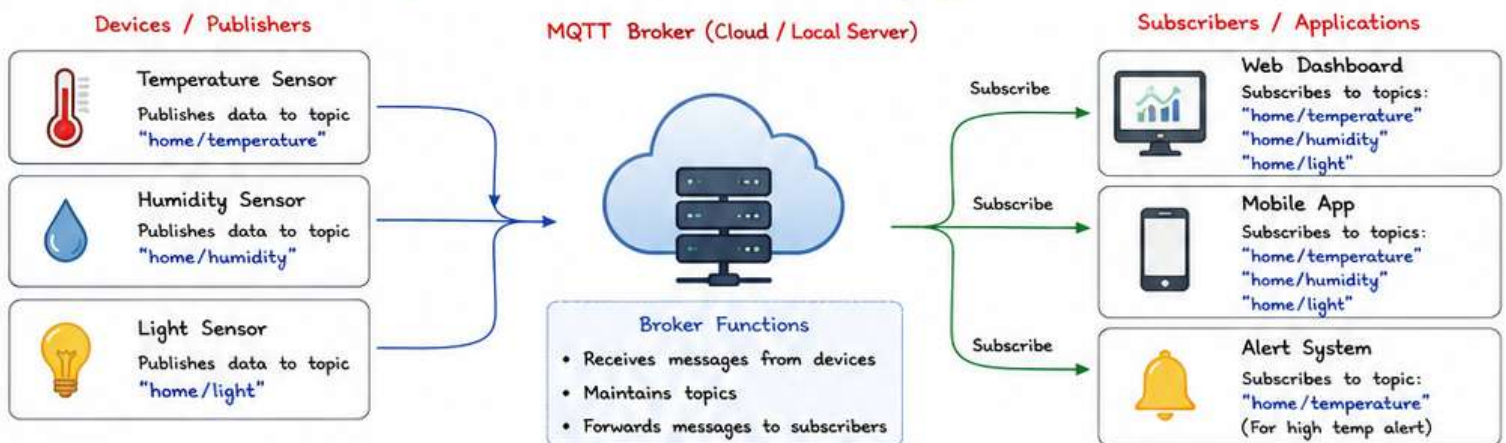
Main Elements :

- **Client** : Any device or application
- **Broker** : Central server
- **Topic** : A named channel
- **Message** : Data published to a topic

5) Quality of Service (QoS) Levels in MQTT

QoS Level	Name	Description	Use Case
0	At most once	Message is delivered at most once. No acknowledgement.	Sensor data where some loss is acceptable.
1	At least once	Message is delivered at least once. PUBACK is sent.	Important data where duplication can be tolerated.
2	Exactly once	Message is delivered exactly once using 4-way handshake.	Critical data where no loss and no duplication allowed.

6) Use of MQTT in IoT Application : Smart Home Monitoring System



7) Benefits in Smart Home Application

- Real-time monitoring of home environment
- Low power sensors can send data for long time
- Works over Wi-Fi / cellular with low bandwidth
- Easily scalable (more sensors can be added)
- Reliable data delivery with QoS
- Easy integration with cloud platforms

8) Example Scenario

A temperature sensor publishes data to topic "home/temperature". The MQTT broker receives it and forwards the data to all subscribers like mobile app and dashboard. If temperature is above 35°C, alert system receives the message and sends alert to the user.

Conclusion : MQTT is a lightweight, efficient and scalable protocol ideal for IoT applications. It enables reliable communication between devices and applications using publish/subscribe model. It is widely used in smart homes, smart cities, industrial IoT, agriculture, healthcare and many more.

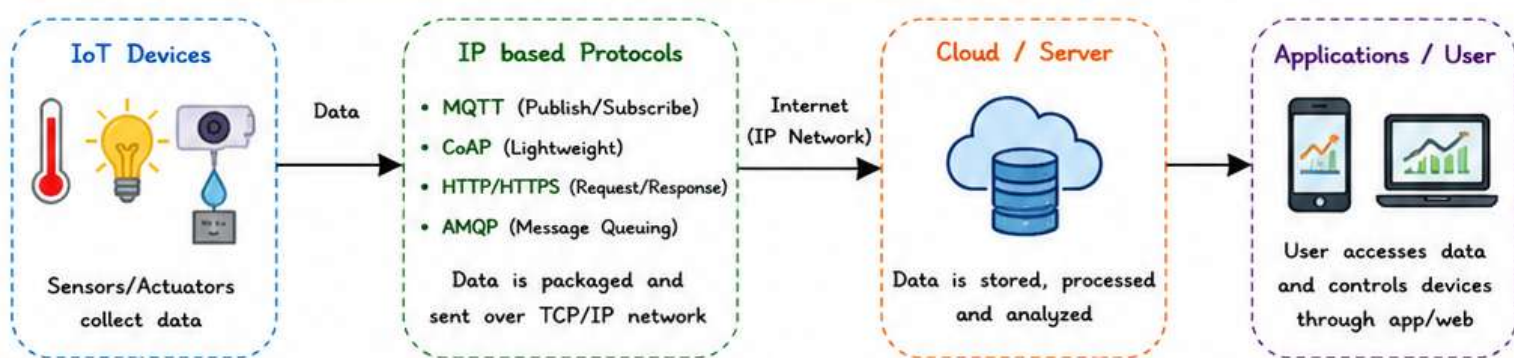
Q10. Illustrate the various IoT applications developed using IP based protocols. [6]

Ans: IP based protocols allow IoT devices to connect to the internet and communicate reliably over TCP/IP network. They are widely used in many IoT applications.

1. Various IoT Applications using IP based protocols

Application	Description	IP based Protocols Used	Example / Benefit
 1. Smart Home Automation	Devices like lights, fans, AC, cameras are controlled and monitored over internet.	HTTP/HTTPS, MQTT, CoAP	Remote control using mobile app, energy saving, security monitoring.
 2. Healthcare Monitoring	Wearable devices and medical sensors send patient data to cloud for monitoring.	MQTT, CoAP, HTTP/HTTPS	Real-time health tracking, alerts to doctors, better patient care.
 3. Industrial IoT (IIoT)	Machines and sensors send data for monitoring, predictive maintenance and control.	MQTT, AMQP, HTTP/HTTPS	Predict faults, reduce downtime, improve productivity.
 4. Smart City Applications	Used in traffic control, smart parking, waste management, street lighting etc.	CoAP, MQTT, HTTP/HTTPS	Better city management, resource saving, improved citizen services.
 5. Smart Agriculture	Sensors monitor soil, weather, irrigation and crop health and send data to cloud.	MQTT, CoAP, HTTP/HTTPS	Increase crop yield, optimize water use, reduce manual work.
 6. Environmental Monitoring	Sensors measure air quality, temperature, humidity and send data for analysis.	MQTT, HTTP/HTTPS, CoAP	Pollution monitoring, early warning systems, climate data collection.

2. Use of IP based protocols in IoT Applications (How it works)



Examples of Protocol Use in Real Applications

- **Smart Home:** ESP32 camera uses HTTP to send video to cloud and mobile app.
- **Health Monitoring:** Wearable device publishes heart rate using MQTT to broker; doctor app subscribes.
- **Smart Agriculture:** Soil sensor sends moisture data using CoAP to cloud server; app shows on dashboard.
- **Industrial IoT:** Machine sensors send data using MQTT to server; analytics predicts machine failure.
- **Smart City:** Street light controller sends status using HTTP/HTTPS to central server for monitoring.



Conclusion: IP based protocols provide standard, scalable and reliable way for IoT devices to connect over internet and are used in wide range of real world applications.

Q8. Show with suitable reasons why Zigbee is popular than Wi-Fi and Bluetooth in IoT. [5]

Ans: Zigbee is more popular in IoT applications than Wi-Fi and Bluetooth because it is low cost, low power, reliable and made for IoT.

Feature	Zigbee	Wi-Fi	Bluetooth
Power Consumption	Very low power. Works for months/years on small battery. 	High power. Consumes more energy, not suitable for battery devices. 	Medium power. Better than Wi-Fi but higher than Zigbee. 
Cost	Low cost chips and modules. Good for large scale deployments. 	High cost modules. More expensive for small IoT devices. 	Moderate cost. Higher than Zigbee in some cases. 
Network Type	Mesh network. Devices talk to each other and extend range. 	Star network. All devices connect to router. If router fails, network down. 	Point to point. One to one connection. Not suitable for large networks. 
Range	Long range (10 - 100 m per hop). Mesh extends coverage further. 	Medium range (20 - 50 m indoors). Limited by router. 	Short range (5 - 10 m). Limited coverage. 
Data Rate	Low data rate (250 kbps). Enough for sensors and control. 	High data rate (Mbps to Gbps). More than required for IoT. 	Low to medium data rate (up to 3 Mbps). Less than Wi-Fi. 
Best Use	Built for IoT. Best for sensors, home automation, smart energy, industrial IoT. 	Good for internet access, video streaming, high data applications. 	Good for connecting mobile accessories, audio devices, small file transfer. 



Conclusion: Zigbee is popular in IoT because it is low power, low cost, supports mesh networking, has long battery life and is designed for large number of devices.

Q9. Critically evaluate the advantages and limitations of Zigbee technology in home automation systems. [5]

Ans:

Advantages of Zigbee in Home Automation	Limitations of Zigbee in Home Automation
Low Power Consumption: Devices run for long time on small batteries. Ideal for sensors and switches.  Low Cost: Zigbee modules are cheap. Good for large number of devices in home.  Mesh Networking: Devices connect with each other. If one device fails, network still works.  Reliable and Stable: Good performance indoors with less interference.  Scalable: Supports many devices (up to 65,000). Easy to add more devices in future.  Designed for IoT: Perfect for small data applications like lighting, temperature, security, energy monitoring. 	Low Data Rate: Only up to 250 kbps. Not suitable for video or high data applications.  Limited Range per Hop: 10 - 100 m per hop. Needs more devices for long distance.  More Complex Setup: Mesh network configuration is more complex than Wi-Fi.  Interference: Works on 2.4 GHz band. Can be affected by Wi-Fi, Bluetooth, microwave ovens.  Requires Hub/Gateway: Needs a Zigbee coordinator or gateway to connect with internet or mobile apps.  Not Suitable for High Data Devices: Not good for devices like cameras, smart TVs, gaming etc. 



Conclusion: Zigbee is excellent for low power, low cost and reliable home automation applications. But it has low data rate and needs proper network planning.

Q11. BLUETOOTH



Explain Bluetooth technology in detail. Describe its architecture, working, physical layer, packet (header) format, types, applications, advantages and limitations.

1. INTRODUCTION

- Bluetooth is a short range wireless communication technology used to exchange data between fixed or mobile devices over short distances.
- It works in the ISM band of 2.4 GHz.
- It is defined by Bluetooth Special Interest Group (SIG).
- It is low power, low cost and easy to use.

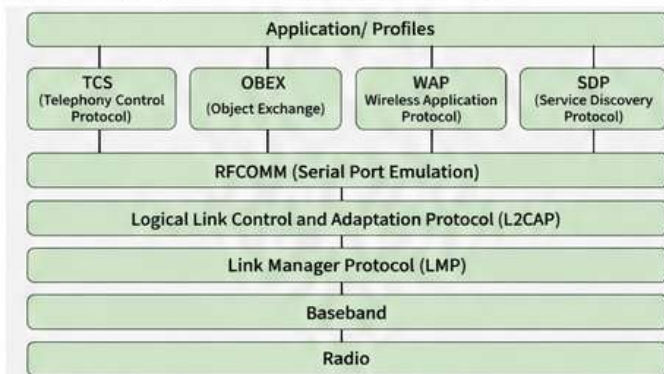
2. FEATURES

- Short range: 1 to 100 meters (Class dependent)
- Low power consumption
- Low cost, small size
- Supports point-to-point and point-to-multipoint
- Uses frequency hopping spread spectrum (FHSS)
- Supports voice and data

3. BLUETOOTH ARCHITECTURE (PROTOCOL STACK)

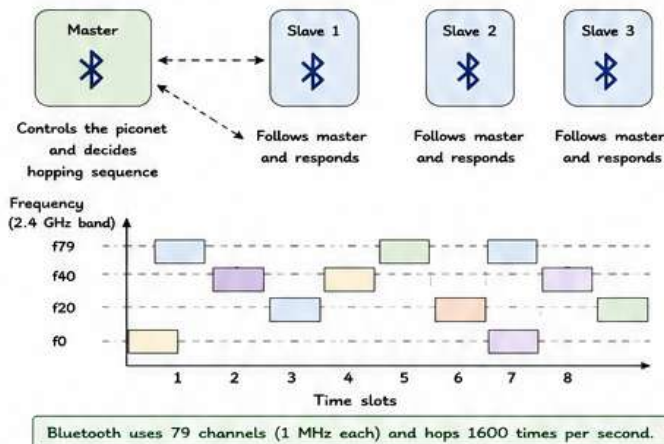
Bluetooth Protocol Stack

The Bluetooth protocol stack consists of multiple layers, each responsible for specific functions to enable reliable communication between devices.



4. WORKING

1. **Device Discovery:** Devices scan for other Bluetooth devices.
2. **Pairing:** Devices authenticate and create a link.
3. **Connection:** Logical link is established (piconet formation).
4. **Data Transfer:** Data is transmitted on ACL (Asynchronous Connection Less) or SCO (Synchronous Connection Oriented) links.
5. **Hopping:** Data is sent using frequency hopping to avoid interference.

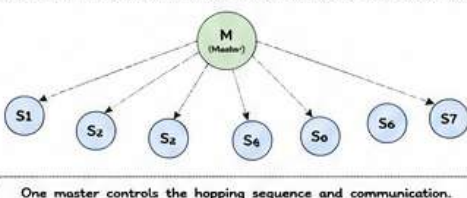


5. PHYSICAL LAYER

- Frequency Band : 2.400 GHz to 2.4835 GHz (ISM band)
- Channels : 79 channels, 1 MHz spacing
- Modulation : GFSK (Gaussian Frequency Shift Keying)
- Data Rate : 1 Mbps (Basic Rate) and 2/3 Mbps (EDR)
- Technique : Frequency Hopping Spread Spectrum (FHSS)

12. PICONET (Diagram)

A piconet is a network formed by one master and up to seven active slaves.



6. PACKET (HEADER) FORMAT

72 bits	54 bits	0-2745 bits	16 bits
Access Code	Header	Payload	FCS
Sync word (0x8E89BED6) for timing and frequency synchronization	Contains Address, Type, Flow, ARQN, SEQN, Length etc.	User data (variable length)	16-bit CRC for error checking

Header Fields (54 bits):

- AM_ADDR (3 bits) : Active Member Address
- TYPE (4 bits) : Type of packet
- FLOW (1 bit) : Flow control
- ARQN (1 bit) : Acknowledgement request
- SEQN (1 bit) : Sequence number
- HEC (8 bits) : Header error check
- Length (16 bits) : Payload length
- L_ADDR (3 bits) : Logical address
- CRC (16 bits) : Error check for payload

7. TYPES OF BLUETOOTH DEVICES (BY RANGE/POWER)

Class	Max Output Power	Range (Approx.)	Applications
Class 1	100 mW (20 dBm)	Up to 100 m	Industrial, outdoor applications
Class 2	2.5 mW (4 dBm)	Up to 10 m	Mobile phones, headsets
Class 3	1 mW (0 dBm)	Up to 1 m	Wearables, sensors, small devices

8. BLUETOOTH PROFILES (Examples)

Profile	Use
SPP (Serial Port Profile)	Serial cable replacement
HFP (Hands Free Profile)	Hands free devices, car kits
A2DP (Advanced Audio Distribution Profile)	High quality stereo audio
HID (Human Interface Device)	Keyboard, mouse, joystick
PAN (Personal Area Network)	Internet access, networking

9. APPLICATIONS

- Wireless headsets and earphones
- Data transfer between mobile phones
- Wireless keyboards, mouse, game controllers
- Smart watches and fitness bands
- Home automation and IoT devices
- Industrial automation and medical devices



10. ADVANTAGES

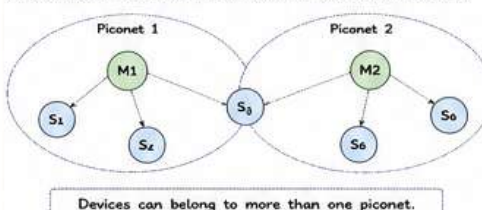
- ✓ Low power consumption
- ✓ Low cost and simple to implement
- ✓ Easy to use and supports plug-and-play
- ✓ Supports multiple devices (piconet and scatternet)
- ✓ Secure (authentication and encryption)
- ✓ Works in license free 2.4 GHz band

11. LIMITATIONS

- ✗ Short range compared to Wi-Fi
- ✗ Lower data rate (less than Wi-Fi)
- ✗ Interference with other 2.4 GHz devices (Wi-Fi, microwave)
- ✗ Not suitable for large file transfer or high bandwidth apps
- ✗ Limited number of active connections in piconet (up to 7 slaves)

13. SCATTERNET (Diagram)

A scatternet is formed by interconnection of two or more piconets.



14. CONCLUSION

Bluetooth is a short range, low power, low cost and easy to use wireless technology. It uses FHSS in 2.4 GHz band and supports voice and data. It is widely used in mobile devices, wearables, home automation and IoT applications.



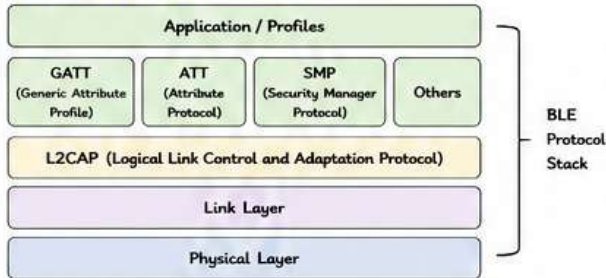
A. BLUETOOTH LOW ENERGY (BLE)

BLE (Bluetooth Low Energy) is a wireless communication technology designed for low power consumption, short range and low data rate applications. It is part of the Bluetooth 4.0 and later versions.

1. FEATURES

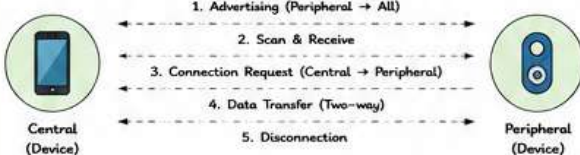
- Very low power consumption
- Short range: 10 – 100 meters
- Low data rate (up to 2 Mbps)
- Operates in 2.4 GHz ISM band
- Small size and low cost
- Supports many devices (central can connect to multiple peripherals)
- Uses frequency hopping for reliability
- Secure with encryption

2. ARCHITECTURE



3. WORKING

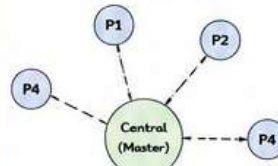
1. **Advertising:** Peripheral device broadcasts advertising packets.
2. **Scanning:** Central device scans and receives advertisements.
3. **Connection:** Central sends connection request, link is established.
4. **Data Transfer:** Data is exchanged using GATT services and characteristics.
5. **Disconnection:** Link is terminated when communication is over.



4. BLE TOPOLOGIES

1) PICONET (Star Topology)

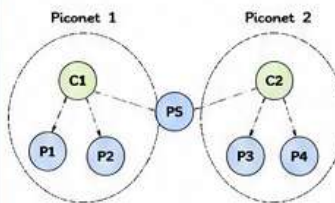
One central device communicates with multiple peripheral devices.



P = Peripheral

2) SCATTERNET (Multi-Piconet)

Two or more piconets are connected through a common device.



C = Central (Master)
P = Peripheral (Slave)

5. PHY LAYER (BLE)

- Frequency Band : 2.4 GHz ISM Band
- Channels : 40 channels (2 MHz spacing)
- Modulation : GFSK (Gaussian Frequency Shift Keying)
- Data Rate : 1 Mbps, 2 Mbps (BLE 5)
- Tx Power : -20 dBm to +10 dBm (typical)
- Range : 10 – 100 meters (class dependent)

6. BLE ADVANTAGES

- ✓ Low power – suitable for battery devices
- ✓ Low cost and simple to implement
- ✓ Fast connection and low latency
- ✓ Supports many devices
- ✓ Secure communication
- ✓ Works with smartphones and IoT devices

7. BLE LIMITATIONS

- ✗ Short range compared to some other technologies
- ✗ Low data rate – not suitable for audio/video streaming
- ✗ Interference in 2.4 GHz band
- ✗ Not ideal for large scale mesh networking

8. BLE APPLICATIONS

- Wearable devices (fitness bands, smart watches)
- Health care and medical devices
- Home automation and smart lighting
- Beacons and location tracking
- Industrial sensors and monitoring
- IoT devices and smart toys

9. BLE VERSIONS (SUMMARY)

Version	Year	Key Features
4.0	2010	Introduced BLE
4.1	2013	Better co-existence, Speed
4.2	2014	Privacy, IPv6 support
5.0	2016	2 Mbps, Long range, More data
5.1	2019	Direction finding, AoA/AoD
5.2	2020	LE Audio, Isochronous channels
5.3	2021	Better reliability, power control

B. Z-WAVE

Z-Wave is a wireless communication protocol designed for home automation and smart homes. It is a low power, low data rate technology that uses mesh networking to provide reliable communication between devices.

1. FEATURES

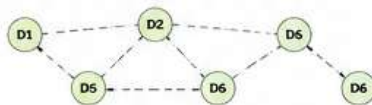
- Operates in sub-GHz frequency band (less interference)
- Low power consumption
- Mesh network for wide coverage and reliability
- Low data rate (up to 100 kbps)
- Secure (AES-128 encryption)
- Interoperable (certified Z-Wave devices work together)
- Supports many devices (up to 232 per network)

2. ARCHITECTURE (PROTOCOL STACK)



3. WORKING

1. **Inclusion:** New device is added to the Z-Wave network.
2. **Routing:** Devices form a mesh network and find the best route.
3. **Communication:** Data is sent hop-by-hop between devices.
4. **Acknowledgement:** Each hop acknowledges for reliability.
5. **Exclusion:** Device can be removed from the network.

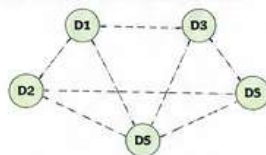


D = Z-Wave Device
All devices act as repeater to extend the network.

4. Z-WAVE TOPOLOGIES

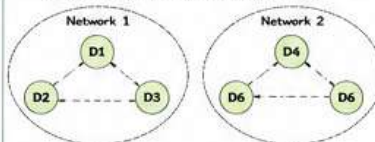
1) PICONET (Mesh Network)

All devices are connected in a mesh. Any device can communicate with any other device through intermediate nodes.



2) SCATTERNET (Multiple Networks)

Two or more Z-Wave networks operate in the same area without interference.



5. PHY LAYER (Z-WAVE)

- Frequency Band : Sub-GHz (varies by region)
-868.4 MHz (EU), 908.4 MHz (US), 921.4 MHz (ANZ)
- Channels : 1 channel
- Modulation : GFSK
- Data Rate : 9.6 kbps – 100 kbps
- Range : Up to 30 meters (indoor), more with mesh
- Tx Power : Up to +14 dBm (varies by region)

6. Z-WAVE ADVANTAGES

- ✓ Reliable mesh network – wide coverage
- ✓ Low power – battery can last for years
- ✓ Works well in smart home environments
- ✓ Less interference (uses sub-GHz band)
- ✓ Interoperable certified devices
- ✓ Secure communication

7. Z-WAVE LIMITATIONS

- ✗ Low data rate – not suitable for high bandwidth apps
- ✗ Requires Z-Wave certified devices (cost higher)
- ✗ Less popular than Wi-Fi or BLE in some regions
- ✗ Range without mesh is limited

8. Z-WAVE APPLICATIONS



9. Z-WAVE GENERATIONS

Generation	Year	Key Features
Z-Wave	2005	9.6 kbps, basic features
Z-Wave Plus	2010	100 kbps, better range, security
Z-Wave 700	2018	Better performance, 500 series chips
Z-Wave Long Range	2021	Up to 1.6 km line-of-sight, low power

COMPARISON SUMMARY

Feature	BLE	Z-Wave
Frequency	2.4 GHz	Sub-GHz (868/908/921 MHz)
Range	10 – 100 m	Up to 30 m (mesh: wider)
Data Rate	Up to 2 Mbps	Up to 100 kbps
Power	Very Low	Low
Network	Star (Piconet), Scatternet	Mesh
Applications	Wearables, IoT, Beacons	Home Automation, Smart Home



Quick Notes

- BLE is best for low power personal area networks and wearables.
- Z-Wave is best for smart home automation and reliable mesh networks.